

STAGE 1: Task Mastery (English VoQA)

 3.35M English VoQA
Photos
(Watermarked)

Extract Text

Train

STAGE 2: Multilingual Deep Supervision

 SeamlessM4T
(Translates Q's & A's to
target language)

Give Translated Text

 PIL + Noto Fonts
(Re-renders watermarks on
physical pixels)

Generate

Generate

 Finnish VoQA Photos
'Onko muki?' on pixels

 Italian VoQA Photos
'C'è una tazza?' on pixels

Train

Train

STAGE 3: Composable Language Adapters

 Finnish Language
Adapter
(LoRA Rank 4)

 Italian Language
Adapter
(LoRA Rank 4)

Parrot Concept:
The Language Adapter
conditions
the visual tokens so they
don't
pivot to English.

MLM / LoRA LEGO Concept:
Task math and Language
math
live in perfectly separate
LoRA channels.
No catastrophic forgetting!

Plugs On Top Of

Plugs On Top Of

 Task Adapter (LoRA Rank
8)
Goal: Master OCR & 8 SFT
structures

Plugs Into

 InternVL-1B / Qwen-2B
(Frozen Core Pre-trained
Brain)